CSC/SDS 109: Communicating with Data

A5: Physical Visualization—Dining Purchase Data

Goals:

- Find interesting trends or patterns in the dining purchase data
- Compose an interactive, physical visualization of the dining purchase data

Important Notes:

- Notice that part 1 of this assignment spans 2 weeks. The quality of what you produce should reflect two full weeks of work; do your best to deliver visualization(s) that are creative, precise, and meaningful.
- You may work alone or in small groups (2-3). If you choose to work in a group, the quantity and quality of your visualization(s) should scale accordingly.
- There are many resources on campus that you can utilize to help build your physical visualization. The Design Thinking Initiative's (https://www.smith.edu/academics/integrative-learning/centers-initiatives/design-thinking/about-dti) map can help you find them-- https://smithmakersmap.com/.

Instructions – Part 1 DUE IN CLASS ON VIEWING DAY

For this assignment, you will analyze data from CEEDS on Smith College Dining Purchases. You can find the inks to the relevant dataset on the course website under "Assignments":

- **Dining_Purchasing_FY2023.xlsx**: contains purchasing information for FY 2023. There is a lot of data in this workbook including purchasing totals by food category, source, and other special designations such as sustainable, local, fair, humane.

Your task for this assignment is to create an interactive physical visualization that enables a viewer to learn about or explore Smith's dining purchases. Your visualization must be interactive and invite the viewer to add their own data to the display. It must also have a clear purpose. That purpose might be to communicate a specific analytical insight, to motivate action from the viewer, to teach the viewer about purchasing practices, etc.; you choose.

Start by exploring the dataset. What does it capture? How do the sheets in the workbook relate? What questions do you have about the data? What questions could the data help you answer? You might want to stop by CEEDS and see if anyone there wants to talk about the data. (They might even have ideas for visuals that would be meaningful or helpful to them!)

Competency Demonstration – Part 2, start in class on viewing day

On viewing day (see course calendar), bring your complete visualization to class. Viewing day will be reserved to engage with each other's visualizations and provide feedback. You will also get verbal feedback from the instructor on viewing day. One way to provide feedback and demonstrate competencies is to write a thoughtful and actionable critique of someone else's visualization(s). If you choose to do this, be sure to cite visualization science and theories to support your suggested changes and compliments.

This is a competency-based course. After you complete your assignment, write a short, succinct, and clear reflection explaining which core competencies you demonstrated in your work and how.

It is recommended that you use this assignment to demonstrate 2-3 of the following competencies:

- Map data to appropriate visual channels
- Accurately encode data values via color
- Apply perceptual concepts to visualization design
- Appropriately use visual embellishment
- Create a data story using storytelling techniques
- Design and implement an accessible visualization
- Write a thoughtful and actionable critique of someone else's work
- Redesign a visualization based on critiques
- Positively contribute to the classroom community

If you choose to demonstrate "Write a thoughtful and actionable critique of someone else's work", include your critique and an image of the visualization it critiques in your submission.

If you choose to demonstrate "Redesign a visualization based on critiques", include the critiques you received and a clear, concise, detailed sketch/outline of how you would revise your visualization based on the critiques.

Submission

Create a PDF with images of your visualization(s) and your write up from part 2. Submit your completed work as a PDF on Gradescope.